

ViZeroLJP: A Retrieval-Augmented Legal Reasoning Framework for Zero-shot Vietnamese Legal Judgment Prediction

Hung Pham-Thanh Nguyen^{1,2}, Son T. Luu^{1,2}, and Kiet Van Nguyen^{1,2,*}

¹Faculty of Information Science and Engineering, University of Information Technology, Ho Chi Minh City

²Vietnam National University, Ho Chi Minh City, Vietnam

Email: hungpt.19@grad.uit.edu.vn, sonlt@uit.edu.vn, kietmv@uit.edu.vn

September 13, 2026

Abstract

This paper presents ViZeroLJP, a zero-shot framework for Vietnamese Legal Judgment Prediction. The framework takes a case fact as input, retrieves relevant legal provisions, and predicts the final judgment label without task-specific fine-tuning. This draft currently focuses on the methodology and experimental protocol.

Keywords: Vietnamese legal NLP, legal judgment prediction, zero-shot learning, law retrieval, large language models

1. Introduction

Vietnamese courts are facing increasing pressure from a growing and more complex caseload. According to the 2025 report of the Supreme People’s Court of Vietnam presented to the National Assembly, courts handled 683,341 cases and resolved 618,341 cases in 2025, while the number of cases continued to increase and the requirements on speed and quality became higher [11]. This context makes computational support for legal work increasingly important. Systems that can help read long case facts, retrieve relevant legal provisions, extract legally decisive signals, and support outcome-oriented reasoning can reduce repetitive analytical workload and provide useful assistance to legal practitioners.

Legal Judgment Prediction (LJP) is the task of predicting the judicial outcome of a case from its textual description and relevant legal context. In this work, the input is the full Vietnamese case fact text, and the output is one of four labels: the plaintiff fully wins, the plaintiff partially wins with more than 50% of the claims accepted, the plaintiff partially wins with 50% or less accepted, or the plaintiff fully loses. LJP has been studied extensively in other jurisdictions through large benchmarks and specialized methods. CAIL2018 introduced a large-scale Chinese judgment prediction dataset [27], Swiss-Judgment-Prediction provided a multilingual and diachronic benchmark for Swiss cases [22], and NyayaAnumana expanded LJP to the Indian legal context [21]. Methodologically, prior work has explored prompt learning with legal knowledge [25], case-law retrieval for outcome prediction [2], legal prompt engineering [26], and logic-enhanced LLM reasoning for difficult cases [28].

In Vietnamese legal NLP, existing research has made progress in legal question answering, legal information retrieval, and legal entailment. Early Vietnamese legal QA work formulated answering legal questions as article-level legal retrieval [9]. R2GQA introduced a retriever-reader-generator system for higher-education regulations [6], while VLQA provides a large expert-annotated dataset for Vietnamese legal article retrieval and question answering [19]. Other studies have developed knowledge-infused retrieval for Vietnamese labor law [23] and natural language inference resources for Vietnamese legal texts [7]. However, compared with legal QA and retrieval, Vietnamese LJP remains underexplored. The task is especially difficult because Vietnamese judgments are long, fact-dense, and contain multiple functional sections, including plaintiff claims, defendant arguments, procuracy opinions, court reasoning, and operative decisions.

To address this gap, we propose ViZeroLJP, a zero-shot retrieval-augmented framework for Vietnamese legal judgment prediction. The framework decomposes the task into three modules. First, an input-processing module compresses a long case fact into structured legal signals, including claims, outcome cues, and candidate decisive text spans. Second, a law-retrieval module constructs a case-aware legal query and retrieves relevant statutory provisions from the public law corpus. Third, a structured reasoning module uses the processed case representation and retrieved provisions to predict the final outcome label. The framework does not require training or fine-tuning the underlying language model on LJP examples, making it suitable for zero-shot and resource-constrained Vietnamese legal settings.

Experiments on the ALQAC 2026 public benchmark show that ViZeroLJP improves over prompt-only inference across multiple open-source backbones. Ablation results further indicate that both input processing and law retrieval contribute to the final performance. Overall, these results suggest that a lightweight zero-shot framework combining case structuring, legal retrieval, and structured reasoning can improve Vietnamese LJP without task-specific model training.

2. Related Work

2.1. Vietnamese Legal NLP

Vietnamese legal NLP has mainly evolved around legal question answering, legal information retrieval, legal textual entailment, and domain-specific legal knowledge representation. Kien et al. [9] formulate Vietnamese legal QA as article-level legal article retrieval and propose a neural attentive text representation model evaluated on an annotated corpus of 5,922 Vietnamese legal questions. More recent legal QA resources include R2GQA, a retriever-reader-generator system for higher-education legal regulations paired with the ViRHE4QA dataset of 9,758 question-answer pairs [6], and VLQA, a large expert-annotated Vietnamese legal QA dataset designed for both legal article retrieval and question answering [19].

Another line of work focuses on retrieving and structuring Vietnamese legal documents. Pham et al. [23] propose a knowledge-infused retrieval framework for Vietnamese labor law, constructing large-scale legal document and Q&A resources and injecting ontology-based knowledge into retrieval models. Related ontology-based studies build intelligent search systems for labor law [17] and Legal-Onto representations for tracking updates in Vietnamese land law documents [5]. These works show that retrieval quality can benefit from legal structure, document hierarchy, and domain knowledge.

Vietnamese legal entailment has also started to receive attention. Nguyen et al. [18] integrate pre-trained language models with an SVM-inspired layer for the VLSP 2023 Legal Textual Entailment Recognition task, while ViLegalNLI [7] introduces a large-scale Vietnamese legal natural language inference benchmark built from statutory documents. In parallel, Vietnamese research groups have explored legal retrieval and entailment with large language models in COLIEE-style settings [14, 15], showing the broader potential of LLM-based legal information processing.

Despite these advances, existing Vietnamese legal NLP resources still concentrate mostly on natural-language questions, statute retrieval, QA, or premise-hypothesis entailment. The inputs in these tasks are usually short questions, legal statements, or controlled legal passages. They do not fully address court-oriented decision-support scenarios in which the input may be a long, noisy, and factually complex case narrative. Such scenarios require additional capabilities, including extracting legally decisive facts from case content, constructing case-aware legal queries, retrieving relevant provisions, and predicting judicial outcomes.

2.2. Legal Judgment Prediction

Legal Judgment Prediction (LJP) aims to predict judicial outcomes from case materials, such as fact descriptions, claims, statutes, precedents, or other procedural information. Early large-scale benchmarks framed LJP largely as a supervised text classification problem. CAIL2018 [27] introduced a large Chinese criminal-case dataset with more than 2.6 mil-

lion cases and multiple prediction targets, including applicable law articles, charges, and prison terms. Swiss-Judgment-Prediction [22] extended the benchmark landscape to a multilingual and diachronic setting with 85K cases from the Federal Supreme Court of Switzerland, showing that text length, legal area, year, and jurisdictional factors affect prediction performance. More recently, NyayaAnumana [21] provided a large Indian LJP corpus and introduced INLegalLlama, a domain-specialized LLM trained with continual pretraining and task-specific supervised fine-tuning.

Modeling long and complex legal documents remains a central challenge in LJP. Several studies therefore move beyond flat text classification by introducing hierarchical encoders, long-context transformers, explanation extraction, or legal knowledge injection. KnowPrompt4LJP [25] aligns Chinese LJP with prompt learning and incorporates external legal keywords into the prompt, showing strong zero-shot and few-shot performance on CAIL2018. Maqsood et al. [10] propose a hierarchical transformer-based architecture for both judgment prediction and explanation extraction, highlighting the importance of explainability in practical legal AI systems. RLJP [28] further argues that LJP needs explicit legal reasoning logic and proposes first-order-logic-enhanced rules optimized through comparative learning for difficult and confusable cases.

With the emergence of large language models, LJP has increasingly been studied under prompting, retrieval-augmented, and realistic decision-support settings. Trautmann et al. [26] investigate zero-shot legal prompt engineering on long case texts from the European Court of Human Rights and the Swiss Federal Supreme Court, showing that general-purpose LLMs can be transferred to LJP without fine-tuning but still lag behind supervised systems. Shui et al. [24] conduct a systematic evaluation of LLMs for LJP and show that similar-case retrieval and label candidates can help LLMs recall useful legal knowledge, although weak LLMs may fail to benefit fully from strong retrieval modules. Nigam et al. [20] emphasize a more realistic setup in which only information available before decision-making is used, and they report that statutes, precedents, and arguments improve LJP while current LLMs remain below expert-level performance. PILOT [2] studies LJP in case-law systems and demonstrates the importance of retrieving relevant precedent cases and modeling temporal shifts in legal principles. In Japanese civil tort cases, Kadowaki and Kano [8] show that even when a fine-tuned ModernBERT system performs competitively on structured tort prediction and rationale extraction, extracting the necessary information directly from raw judgment documents remains substantially more difficult.

Overall, prior LJP research has established large-scale benchmarks and strong supervised, prompt-based, and retrieval-augmented methods across Chinese, European, Swiss, Indian, Japanese, and common-law settings. However, these studies also reveal persistent challenges: legal documents are long, outcome labels may depend on subtle

claim-level distinctions, relevant statutes or precedents are often required for grounded reasoning, and realistic prediction must avoid relying on hindsight signals. For Vietnamese law, public research on LJP remains limited compared with legal QA and legal information retrieval, which leaves open the question of how to perform zero-shot judgment prediction from long Vietnamese case facts while still grounding model reasoning in retrieved legal provisions. ALQAC 2026 has begun to study this task more systematically in the Vietnamese language and legal context. In this work, we focus on addressing Vietnamese LJP under a zero-shot setting, without training or fine-tuning the underlying language model for the target task [1].

3. Methodology

Problem Definition. We study Legal Judgment Prediction (LJP). The input is the full fact content of a legal case:

$$x_i = \text{case_fact}_i, \quad (1)$$

where x_i denotes the complete textual fact field provided for the i -th case, including the plaintiff’s claims, factual background, party statements, related-party statements, and, when available, the procuracy’s opinion.

Given an unlabeled case x_i , the objective is to predict one outcome label $y_i \in \mathcal{Y}$, where $\mathcal{Y} = \{\text{A_WIN}, \text{PARTIAL_A_WIN}, \text{PARTIAL_B_WIN}, \text{B_WIN}\}$. The label A_WIN denotes that the court fully accepts all of the plaintiff’s claims, PARTIAL_A_WIN denotes that the court partially accepts the plaintiff’s claims and the accepted portion is greater than 50%, PARTIAL_B_WIN denotes that the court partially accepts the plaintiff’s claims but the accepted portion is 50% or less, and B_WIN denotes that the court fully rejects all of the plaintiff’s claims. Formally, ViZeroLJP implements a zero-shot predictor

$$f : x_i, \mathcal{L} \rightarrow \hat{y}_i, \quad \hat{y}_i \in \mathcal{Y}, \quad (2)$$

where $\mathcal{L}$ is an external corpus of legal provisions. No task-specific supervised fine-tuning or labeled examples are used during inference.

Method Overview. ViZeroLJP is a retrieval-augmented, hybrid reasoning framework consisting of three sequential modules: (i) input processing, (ii) legal context retrieval, and (iii) outcome inference. Figure 1 illustrates the overall architecture. The first module converts the long case fact into structured case signals. The second module builds a legal query from the processed case and retrieves relevant law articles with BM25. The final module combines claim-response signals, procuracy-guided cues, retrieved legal context, structured LLM extraction, and deterministic quantitative checks to produce the final label.

3.1. Input Processing

The input processor maps a raw case fact x_i to a structured representation p_i :

$$p_i = g_{\text{input}}(x_i). \quad (3)$$

The system segments the case fact using speaker and discourse markers that frequently appear in Vietnamese civil judgments, such as “nguyen don”, “don khoi kien”, “bi don”, “nguoi co quyen loi”, “dai dien vien kiem sat”, and “kiem sat vien”. Each segment is represented as

$$z_{i,t} = (s_{i,t}, u_{i,t}, c_{i,t}, v_{i,t}), \quad (4)$$

where $s_{i,t}$ is the coarse section, $u_{i,t}$ is the speaker, $c_{i,t}$ is the matched discourse cue, and $v_{i,t}$ is the text span. These fields are heuristic descriptors rather than gold annotations.

From the segmented case, ViZeroLJP extracts the plaintiff’s main claim m_i by prioritizing segments associated with the plaintiff and the factual background, then selecting the final request clause introduced by the cue “yeu cau”. The claim extractor also records quantitative request signals such as monetary amounts and land areas:

$$m_i = (r_i, a_i, d_i), \quad (5)$$

where r_i is the textual request, a_i is the set of requested monetary amounts, and d_i is the set of requested land-area quantities.

Finally, the system collects outcome-signal candidates $E_i = \{e_{i,1}, \dots, e_{i,M}\}$ from spans containing cues such as “de nghi Hoi dong xet xu”, “chap nhan mot phan”, “khong chap nhan yeu cau”, “bac yeu cau”, and “rut yeu cau”. These spans are treated as predictive signals rather than ground-truth dispositions, since the fact description can include recommendations or party positions without revealing the court’s final operative order. Each candidate is assigned a deterministic score based on its section, speaker, and lexical cues; the top eight candidates are retained for downstream reasoning.

3.2. Legal Context Retrieval

The retrieval module provides legal grounding without training an embedding model. The legal corpus is first flattened from documents into article-level units

$$\mathcal{L} = \{\ell_j\}_{j=1}^N, \quad (6)$$

where each ℓ_j stores an article identifier, its source law document, and the article text.

ViZeroLJP builds a law query q_i from three sources: the extracted main claim, sentences in x_i that contain legal-issue terms, and explicit references to legal articles appearing in the case fact. Let S_i be the set of selected legal-issue sentences and R_i be the set of extracted article references. The query is

$$q_i = \text{norm}(m_i^{\text{text}} \parallel S_i \parallel R_i), \quad (7)$$

where $\parallel$ denotes text concatenation and $\text{norm}(\cdot)$ applies whitespace normalization. Legal-issue sentences are selected using a fixed Vietnamese keyword inventory covering common civil-law relations, including contracts, compensation, loans, credit, inheritance, land-use rights, transfer contracts, debts, counterclaims, independent claims, estate division,

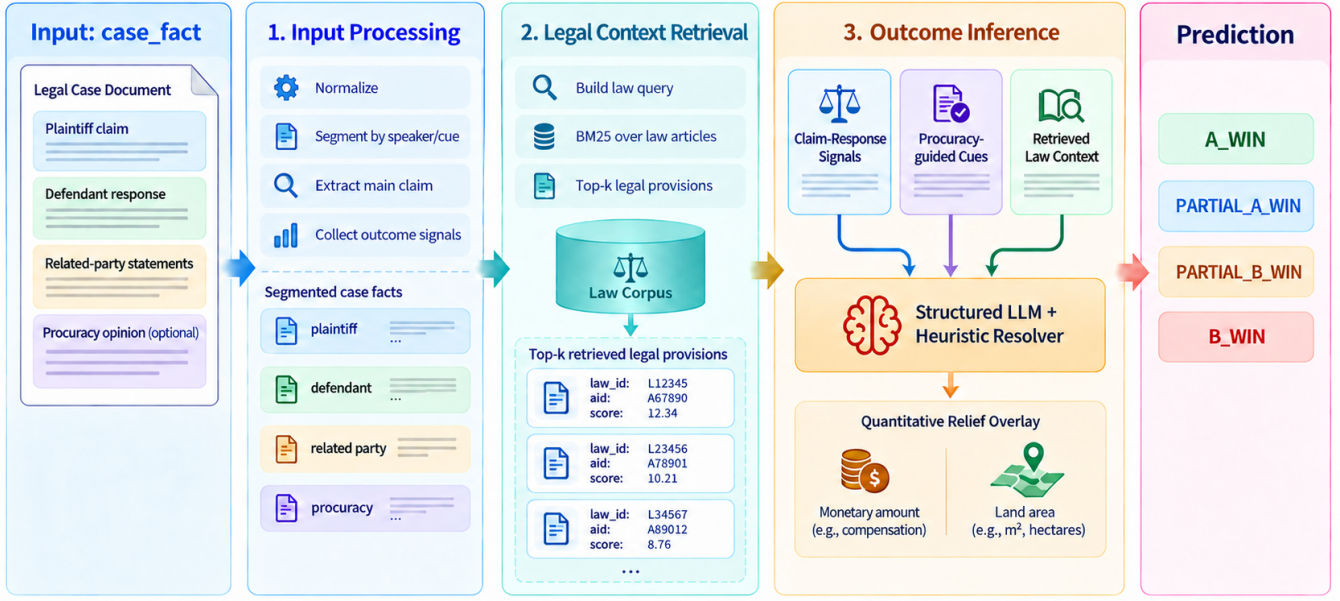


Figure 1: Overview of ViZeroLJP. The system predicts the judgment label from a Legal Case Document, passes it through three modules—Input Processing, Legal Context Retrieval, and Outcome Inference—and outputs one of four prediction labels.

boundaries, and passage disputes. The query is truncated to 6,500 characters.

Retrieval is performed with BM25 over article text. For a query q and article ℓ_j , we write the score in a compact form:

$$\text{BM25}(q, \ell_j) = \sum_{w \in q} \alpha_w \beta_{w,j} \gamma_{w,q}. \quad (8)$$

Here α_w denotes the inverse-document-frequency weight of token w , $\beta_{w,j}$ denotes its length-normalized term-frequency contribution in article ℓ_j , and $\gamma_{w,q}$ denotes its query-frequency contribution. The query-frequency term is capped so that repeated words in a long case fact do not dominate retrieval. The system retrieves the top $K_{\text{law}} = 18$ legal provisions, filters out a small set of non-civil legal documents when enough civil-law candidates remain, and passes at most eight legal evidence passages to the reasoning module.

3.3. Outcome Inference

The outcome inference module predicts the judgment label from the processed case p_i and the retrieved legal context $L_i \subset \mathcal{L}$:

$$\hat{y}_i = g_{\text{infer}}(p_i, L_i). \quad (9)$$

It combines two complementary estimators: a deterministic heuristic resolver and a structured LLM extractor.

Heuristic resolver. The heuristic resolver searches the outcome-signal candidates for legal outcome phrases such as

accepting all claims, accepting part of the claims, rejecting the claims, or withdrawing claims. For each candidate e , the system computes a source-aware score

$$h(e) = \lambda_s(e) + \lambda_c(e) - \delta(e), \quad (10)$$

where λ_s measures source reliability, λ_c weights matched outcome cues, and δ penalizes lower-ranked or weaker matches. Signals from neutral legal authorities, especially the procuracy, receive higher reliability than adversarial party statements, but they are still treated as recommendations rather than final court orders. The best-scoring match is mapped to an intermediate acceptance scope:

$$\text{scope}_i \in \mathcal{S}, \quad \mathcal{S} = \begin{cases} \text{ALL,} \\ \text{MAJORITY,} \\ \text{MINORITY,} \\ \text{NONE,} \\ \text{UNCLEAR} \end{cases} \quad (11)$$

Structured LLM extraction. When an LLM is available, ViZeroLJP prompts it to produce a schema-constrained assessment rather than a free-form answer. The model receives compact processed-case segments and the retrieved legal evidence passages. Its output must match a strict JSON schema with fields for source quality, accepted-claim scope, confidence, requested relief, granted relief, rejected relief, and

cited evidence chunk identifiers. At most seven evidence chunks are cited in the structured output. The LLM is explicitly instructed not to assume information outside the input case fact and not to convert the procuracy’s recommendation into a court verdict. Invalid JSON outputs are repaired through schema-validation retries.

Estimator selection. Let A_i^H be the heuristic assessment and A_i^L be the structured LLM assessment. ViZeroLJP uses the heuristic assessment by default and replaces it with the LLM assessment only when the LLM confidence is at least as high. With $c_i^L = \text{conf}(A_i^L)$ and $c_i^H = \text{conf}(A_i^H)$, the selected assessment is

$$A_i = \begin{cases} A_i^L, & c_i^L \geq c_i^H, \\ A_i^H, & \text{otherwise.} \end{cases} \quad (12)$$

The selected acceptance scope is then mapped to the four LJP labels:

$$\phi(s) = \begin{cases} \text{A_WIN}, & s = \text{ALL}, \\ \text{PARTIAL_A_WIN}, & s = \text{MAJORITY}, \\ \text{PARTIAL_B_WIN}, & s = \text{MINORITY}, \\ \text{B_WIN}, & s = \text{NONE}. \end{cases} \quad (13)$$

If the scope is unclear, the system falls back to `PARTIAL_A_WIN`, reflecting a conservative prior for partially accepted civil claims.

Quantitative relief overlay. Many civil cases hinge on whether the plaintiff receives the full requested amount or only a fraction of it. Therefore, ViZeroLJP applies a final deterministic overlay for monetary and land-area claims. Let u_i be the requested quantity and v_i be an inferred accepted quantity from reliable outcome signals. The accepted ratio is

$$\rho_i = \frac{v_i}{u_i}. \quad (14)$$

If an initial full-win prediction has $\rho_i < 0.995$, it is demoted to a partial-win label. If a partial land-area award satisfies $\rho_i \leq 0.5$, it is mapped to `PARTIAL_B_WIN`; if the accepted area is approximately equal to the requested area, it may be promoted to `A_WIN`. This overlay is conservative and is applied only when explicit quantities can be extracted.

4. Experiments

4.1. Dataset

We evaluate ViZeroLJP on the ALQAC 2026 public benchmark for Vietnamese legal case outcome prediction. The dataset is provided by the ALQAC 2026 organizers to participating teams for the legal judgment prediction task. ALQAC is a Vietnamese legal AI competition held annually from ALQAC 2021 to ALQAC 2026 [1, 3, 4, 12, 13, 16]; therefore, the data provided to participating teams is a reliable benchmark resource. The processed dataset used in this paper is available from the GitHub repository of this work.

The public test set contains 50 Vietnamese civil judgments. Each original sample contains a case identifier, a long `case_fact`, metadata describing the court and litigant roles, and the gold `verdict_label`. In this benchmark, all cases are civil disputes; party A is the plaintiff and party B is the defendant. The label distribution is imbalanced: 16 cases are labeled `A_WIN`, 19 are labeled `PARTIAL_A_WIN`, 5 are labeled `PARTIAL_B_WIN`, and 10 are labeled `B_WIN`. The public law corpus, `corpus_law_pub.json`, is used only as an external legal source for retrieval. No benchmark labels are used for training, tuning, or selecting demonstrations.

4.2. Dataset Processing

To preserve the zero-shot setting and match the formulation of Vietnamese legal judgment prediction in this paper, we filter each original benchmark record into a minimal tuple (`case_id`, `case_fact`, `verdict_label`). The `case_id` is used only for tracking and evaluation alignment, the full `case_fact` is the only textual input to the prediction system, and the gold `verdict_label` is used only after prediction to compute metrics. All other fields, including court name, case type, party roles, party descriptions, court reasoning, and court verdict text, are removed from the model input.

The law corpus is flattened into article-level records containing a law identifier, an article identifier, and the article content. ViZeroLJP then processes each `case_fact` with the three modules described in Section 3: input processing, legal context retrieval, and outcome inference.

4.3. Evaluation Metrics

We report Accuracy, Macro-F1, per-label F1, and Coverage. Accuracy measures exact agreement with the gold label. Macro-F1 gives equal weight to all four labels and is therefore important for this benchmark because `PARTIAL_B_WIN` and `B_WIN` have fewer examples than the two A-side labels. Per-label F1 exposes which outcome classes are improved or remain difficult. Coverage measures the percentage of cases for which a valid label is produced; it is included because structured generation failures can otherwise be hidden by aggregate accuracy. All metrics are reported as percentages.

4.4. Implementation Details

We choose compared models according to their ability to process Vietnamese legal text and their availability as instruction-following language models. We evaluate four open-source model families: Qwen3.5-4B and Qwen3.5-9B, Llama-3.2-3B-Instruct and Llama-3.1-8B-Instruct, Phi-4-mini-instruct and Phi-4, and DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Qwen-14B. For each model family, we select one smaller model and one larger model to observe the effect of parameter scale on zero-shot legal judgment prediction. We also include GPT-5.6 Sol with high reasoning effort as a closed-source prompt-only reference.

For each open-source backbone, we evaluate two settings.

In the first setting, the model receives a direct prompt over the `case_fact` and predicts one of the four labels; we denote this setting as prompt-only. In the second setting, the same backbone is integrated into ViZeroLJP, where it receives the structured case representation and retrieved legal context before prediction. Each open-source setting is run three times, and the paper reports the mean metrics. The Delta row in Table 1 is computed as the mean ViZeroLJP score minus the mean prompt-only score for the same backbone. GPT-5.6 Sol is evaluated once through the ChatGPT web interface because it is a closed-source system.

All open-source experiments are executed on a machine equipped with an NVIDIA RTX 3090 GPU with 24GB VRAM. Models are served through a local OpenAI-compatible `llama.cpp` endpoint using GGUF quantization. For the retrieval-augmented setting, BM25 retrieves the top 18 candidate law provisions. Among them, at most 8 legal provisions are selected into the final reasoning prompt, and the structured LLM inference step receives at most 7 evidence chunks.

4.5. Quantitative Results

Table 1 summarizes the main comparison. Across most open-source backbones, integrating ViZeroLJP improves both Accuracy and Macro-F1 over prompt-only inference. The largest gains are observed for DeepSeek-R1-Distill-Qwen-7B, which improves by 24.67 Accuracy points and 27.66 Macro-F1 points, followed by Llama-3.2-3B-Instruct, Qwen3.5-4B, and Phi-4-mini-instruct. Qwen3.5-9B + ViZeroLJP obtains the strongest open-source Macro-F1 among the tested models, reaching 43.82, and is the only open-source configuration in the main table that reaches a non-zero PARTIAL_B_WIN F1 of 25.00.

An important observation is that compact models benefit substantially from the proposed framework. Qwen3.5-4B gains 18.00 Accuracy points, Llama-3.2-3B gains 20.00 points, Phi-4-mini-instruct gains 17.33 points, and DeepSeek-R1-Distill-Qwen-7B gains 24.67 points after being integrated with ViZeroLJP. These improvements indicate that the proposed input processing and legal retrieval modules reduce the burden on smaller LLMs: instead of reading a long raw judgment and deciding the label directly, the model receives a compressed representation of the dispute, explicit outcome cues, and a limited set of retrieved legal provisions. This makes ViZeroLJP particularly useful when computational resources only allow compact open-source models.

The results also show that increasing the number of parameters does not guarantee better performance in this zero-shot LJP setting. For example, DeepSeek-R1-Distill-Qwen-14B improves only slightly over its prompt-only baseline, while the 7B variant benefits much more from the proposed framework. Similarly, Phi-4-mini-instruct with ViZeroLJP reaches higher Accuracy than Phi-4 with ViZeroLJP. This suggests that schema following, Vietnamese legal text handling, and compatibility with the structured reasoning pipeline are as important as parameter count. GPT-5.6 Sol (high-thinking), a

frontier-scale closed-source model with substantially higher inference cost than the tested open-source backbones, obtains the best overall result with 54.00 Accuracy and 50.65 Macro-F1. However, this result is not far ahead of Qwen3.5-9B + ViZeroLJP, which reaches 53.33 Accuracy and 43.82 Macro-F1 with only a 9B open-source backbone. This comparison suggests that ViZeroLJP can narrow the gap between compact open-source LLMs and expensive frontier models, thereby substantially reducing the computational resources and deployment cost required for Vietnamese legal judgment prediction.

The B-side labels remain challenging. PARTIAL_B_WIN in particular often has zero F1 for open-source models, because the distinction between PARTIAL_A_WIN and PARTIAL_B_WIN requires comparing requested relief with accepted relief and deciding whether the accepted portion is above or below 50%. This boundary is difficult when the case fact contains multiple monetary claims, land-area quantities, or partially rejected subclaims. Nevertheless, ViZeroLJP generally improves the system by making the main claim and outcome cues explicit before the LLM performs label selection.

4.6. Qualitative Analysis

We inspect two kinds of intermediate outputs: legal-context retrieval and input processing. Table 2 shows a representative case where retrieval helps the model correct a prompt-only error. The case concerns compensation for health and property damage. Direct prompting predicts a full plaintiff win, while ViZeroLJP retrieves relevant compensation provisions and predicts the correct partial-win label.

This retrieved law is relevant because the case concerns compensation for health treatment costs and property damage. By providing a legal provision that defines recoverable health-related damages, the retrieval module gives the model a more grounded context for distinguishing a full plaintiff win from a partial plaintiff win.

Table 3 shows that case_4644 is a reasonable retrieval case because the query explicitly mentions compensation for health damage, treatment costs, and property damage; the selected Civil Code provision directly matches this legal issue. By contrast, case_8732 is an unreasonable retrieval case because the query mainly contains identity and address information of a related party. As a result, BM25 retrieves a land-use planning provision from surface lexical cues, but that provision does not explain the decisive legal outcome of the case. Overall, retrieval is useful when the query captures the legal relation of the dispute, but it remains sensitive to query quality and lexical overlap.

Table 4 illustrates the output of the input processing module. The example shows why explicit extraction is useful for long Vietnamese judgments: the system separates the plaintiff’s requested relief from the amount eventually accepted by the court. This structured view directly supports a partial-win decision instead of requiring the model to reason over the entire raw document at once.

Table 1: Main comparison on the ALQAC 2026 public benchmark. Metrics are percentages. Open-source results are averaged over three runs. “△” denotes the average improvement of ViZeroLJP over the corresponding prompt-only baseline. Within each backbone, the better result between prompt-only and ViZeroLJP is shown in bold.

Backbone	Params	Method	Acc.	Macro-F1	A F1	PA F1	PB F1	B F1	Cov.
Llama-3.2-3B-Instruct	3B	+ Prompt-only	27.33	20.85	22.54	30.04	0.00	30.82	100.00
		+ ViZeroLJP (our)	47.33	31.68	45.69	66.69	0.00	14.34	100.00
		△	+20.00	+10.83	+23.15	+36.65	+0.00	-16.48	+0.00
Llama-3.1-8B-Instruct	8B	+ Prompt-only	38.00	28.78	52.76	40.92	0.00	21.44	100.00
		+ ViZeroLJP (our)	47.33	32.75	48.85	59.61	0.00	22.54	100.00
		△	+9.33	+3.97	-3.91	+18.69	+0.00	+1.10	+0.00
Phi-4-mini-instruct	3.8B	+ Prompt-only	36.00	14.52	0.00	58.07	0.00	0.00	100.00
		+ ViZeroLJP (our)	53.33	38.91	52.17	67.10	0.00	36.36	100.00
		△	+17.33	+24.39	+52.17	+9.03	+0.00	+36.36	+0.00
Phi-4	14B	+ Prompt-only	34.67	25.34	22.22	47.62	18.18	13.33	100.00
		+ ViZeroLJP (our)	46.67	35.37	52.56	59.41	0.00	29.52	100.00
		△	+12.00	+10.03	+30.34	+11.79	-18.18	+16.19	+0.00
DeepSeek-R1-Distill-Qwen-7B	7B	+ Prompt-only	27.33	10.79	0.00	43.15	0.00	0.00	100.00
		+ ViZeroLJP (our)	52.00	38.45	50.68	67.64	0.00	35.48	100.00
		△	+24.67	+27.66	+50.68	+24.49	+0.00	+35.48	+0.00
DeepSeek-R1-Distill-Qwen-14B	14B	+ Prompt-only	46.67	37.69	59.04	44.93	0.00	46.77	100.00
		+ ViZeroLJP (our)	48.67	38.04	52.72	58.35	0.00	41.10	100.00
		△	+2.00	+0.35	-6.32	+13.42	+0.00	-5.67	+0.00
GPT-5.6 Sol (high-thinking)	–	+ Prompt-only	54.00	50.65	80.00	25.00	33.33	64.29	100.00
Qwen3.5-4B	4B	+ Prompt-only	34.00	27.53	38.73	43.48	4.76	23.15	100.00
		+ ViZeroLJP (our)	52.00	37.36	55.50	64.91	0.00	29.05	100.00
		△	+18.00	+9.83	+16.77	+21.43	-4.76	+5.90	+0.00
Qwen3.5-9B	9B	+ Prompt-only	44.00	34.35	55.28	46.82	0.00	35.31	100.00
		+ ViZeroLJP (our)	53.33	43.82	47.83	64.97	25.00	37.50	100.00
		△	+9.33	+9.47	-7.45	+18.15	+25.00	+2.19	+0.00

Table 2: A representative case where legal retrieval supports the correct prediction.

Case summary (case_4644): "Nguyên đơn yêu cầu bị đơn bồi thường chi phí điều trị và thiệt hại tài sản do bị cáo buộc làm đổ cột bê tông hàng rào và làm hỏng điện thoại." Gold label: PARTIAL_A_WIN.
Qwen3.5-9B + Prompt-only: A_WIN.
Retrieved law: "Điều luật của Bộ luật Dân sự về thiệt hại do sức khỏe bị xâm phạm, bao gồm chi phí hợp lý cho việc điều trị, thu nhập thực tế bị mất hoặc bị giảm sút, và chi phí hợp lý cho người chăm sóc trong thời gian điều trị."
Qwen3.5-9B + ViZeroLJP (our): PARTIAL_A_WIN -- matches the gold label.

For case_4101, the input-processing module extracts the key information accurately. It identifies the dispute as a compensation claim caused by an animal-related incident, preserves the requested amounts for vehicle repair and medical expenses, and captures the court’s partial acceptance signal. In particular, the extracted accepted amount and rejected amount are both 3,437,500 VND, which makes the partial-acceptance boundary explicit for the outcome inference module.

4.7. Ablation Study

We perform an ablation study with Qwen3.5-9B to measure the contribution of each module. Table 5 shows that removing input processing causes a large drop from 53.33 to 40.00 Accuracy and from 43.82 to 20.73 Macro-F1. Removing law retrieval also reduces performance, with Accuracy falling to 48.00 and Macro-F1 to 35.05. Removing both modules produces the same weak result as removing input processing alone. These findings indicate that input processing is the central component for making long Vietnamese case facts tractable: it extracts the key evidential signals needed for outcome reasoning and also supplies a focused query for legal-context retrieval. At the same time, law retrieval is not merely an auxiliary component. When the retrieved provision matches the legal issue of the dispute, it provides explicit legal grounding that substantially helps the model perform legal reasoning rather than relying only on surface cues from the raw case fact. The full framework benefits from the interaction between these two modules: input processing identifies the important facts and builds a better law query, while law retrieval supplies the relevant normative context for the final prediction.

Table 3: Qualitative inspection of legal-context retrieval with concrete queries and retrieved-law summaries.

Case	Full generated query	Selected retrieved law
case_4644	yêu cầu Nguyên nhân tranh chấp là do anh D bị cáo buộc đập đổ cột bê tông hàng rào của gia đình vợ chồng anh N, chị C. bị đơn bồi thường thiệt hại sức khỏe, chi phí điều trị và thiệt hại về tài sản (điện thoại bị hỏng).	Quy định của Bộ luật Dân sự về thiệt hại do sức khỏe bị xâm phạm. Điều luật này tóm tắt các khoản có thể bồi thường như chi phí điều trị hợp lý, thu nhập bị mất hoặc giảm sút, và chi phí người chăm sóc.
case_8732	Người có quyền lợi, nghĩa vụ liên quan: Bà Đặng Thị H - sinh năm 1962. Trú tại: Thôn T2, xã Y1, huyện Y2, tỉnh Nam Định.	Quy định của Luật Đất đai về căn cứ lập quy hoạch sử dụng đất cấp huyện. Điều luật này không trực tiếp gắn với yêu cầu chính vì query chủ yếu chứa thông tin nhân thân và địa chỉ.

Table 4: Input-processing example for case_4101. The module converts a long `case_fact` into structured signals used by outcome inference.

Field	Extracted signal
Main claim	Nguyên đơn yêu cầu bị đơn bồi thường thiệt hại do sự cố liên quan đến chó của bị đơn.
Requested relief	6.535.000 đồng tiền sửa xe, 340.000 đồng tiền viện phí, tổng cộng 6.875.000 đồng.
Outcome cue	Tòa buộc bồi thường 3.437.500 đồng và không chấp nhận phần còn lại 3.437.500 đồng.

Table 5: Ablation study with Qwen3.5-9B. Metrics are percentages.

Method	Acc.	Macro-F1
w/o Law Retrieval	48.00	35.05
w/o Input Processing	40.00	20.73
w/o Law Retrieval + w/o Input Processing	40.00	20.73
Full ViZeroLJP Setup	53.33	43.82

5. Limitations and Future Work

Although the quantitative results and qualitative analyses show that ViZeroLJP improves zero-shot Vietnamese Legal Judgment Prediction, several limitations remain. First, the legal retrieval module still relies on BM25, a relatively simple lexical retrieval algorithm. BM25 is reproducible and efficient, but it depends heavily on query quality and lexical overlap. As a result, the retriever does not fully capture the legal context of a case when the generated query misses the core dispute or contains mostly peripheral information such as party identity, address, or procedural details.

Second, cases with B_WIN and PARTIAL_B_WIN labels remain difficult for the system. These labels require the model to determine whether the plaintiff’s claims are fully rejected or only accepted up to 50% of the requested relief. This requires careful comparison between the original claims and the accepted relief, which is challenging when a case contains multiple monetary amounts, land-area quantities, non-monetary requests, or several partially rejected subclaims. More research is needed to improve reasoning over these B-side outcomes.

Third, the ALQAC 2026 benchmark does not model the

temporal evolution of legal provisions, such as amendments, supplements, replacements, or validity periods. Therefore, the current benchmark is suitable for evaluating a fixed legal context at a particular point in time, but it does not test whether a system can reason over time-sensitive changes in Vietnamese law. In addition, the dataset does not provide gold labels for legal-provision retrieval. Consequently, the retrieval module is evaluated mainly through qualitative inspection rather than through a fully supervised retrieval metric.

Fourth, the quantitative results demonstrate that ViZeroLJP can improve prediction accuracy, but they do not fully explain why some smaller models outperform larger models from the same or related model families. This suggests that parameter count alone is not sufficient for selecting the most suitable backbone. Factors such as instruction following, Vietnamese legal-language coverage, structured-output reliability, quantization behavior, and interaction with the retrieval-augmented pipeline may also affect performance.

Fifth, ViZeroLJP remains a zero-shot framework. It does not fine-tune the language model on labeled examples from the target task. While this design makes the method lightweight and reproducible, it also means that reliability in real-world deployment still requires supervision and careful validation. We therefore encourage future systems to combine task-specific fine-tuning with the proposed retrieval and structured-reasoning framework. Finally, the current system retrieves statutory legal provisions only; it has not yet incorporated case-law or precedent retrieval for decision support.

Future work will proceed in three directions. Methodologically, we plan to improve retrieval beyond BM25 by exploring dense representations of cases and legal provisions, including embedding-based retrieval with cosine similarity, hybrid lexical-semantic retrieval, and reranking. We also plan to study retrieval over related cases and precedents, improve reasoning on difficult and complex disputes, and move toward a more reliable multi-agent legal reasoning system that can support practical deployment.

For model selection, future work should analyze why larger models can underperform smaller ones in this benchmark. Instead of assuming that larger parameter counts always lead to better results, model selection should consider both cost and task-specific performance. One pos-

sible direction is to use evolutionary algorithms to optimize backbone selection, inference configuration, and cost-performance trade-offs.

For data, it is necessary to build a labeled legal-retrieval dataset for Vietnamese law, including temporal changes in legal principles and documents such as amendments, supplements, replacements, and validity periods. In parallel, larger Vietnamese legal judgment datasets should be developed for fine-tuning language models on outcome prediction. These data resources should focus on the Vietnamese language and the legal context of Vietnam.

6. Conclusion

In this paper, we propose ViZeroLJP, a retrieval-augmented reasoning framework for zero-shot Vietnamese Legal Judgment Prediction. By decomposing the task into input processing, legal-context retrieval, and structured outcome inference, the framework converts long and noisy Vietnamese case facts into compact legal signals, retrieves relevant statutory provisions, and predicts one of four judgment outcome labels without task-specific fine-tuning. This design enables large language models to reason over both the factual content of a case and an explicit legal context, rather than relying only on direct prompting over the raw document.

Extensive experiments on the ALQAC 2026 public benchmark demonstrate that ViZeroLJP consistently improves the performance of multiple open-source backbones over their prompt-only counterparts. The improvements are especially clear for compact models, showing that the proposed framework can reduce the resource gap between small open-source models and expensive closed-source frontier systems. In particular, Qwen3.5-9B integrated with ViZeroLJP achieves a competitive result compared with GPT-5.6 Sol (high-thinking), while requiring a much smaller open-source backbone. The ablation study further confirms that input processing and law retrieval both contribute to the final performance: input processing extracts the decisive case evidence and supports query construction, while law retrieval provides legal grounding for the final reasoning step. Overall, the results show that ViZeroLJP is an effective and resource-conscious approach for Vietnamese legal judgment prediction in a zero-shot setting.

References

- [1] ALQAC 2026 Organizers. ALQAC 2026. <https://sites.google.com/view/alqac2026>, 2026. Accessed: 2026-09-12.
- [2] Lang Cao, Zifeng Wang, Cao Xiao, and Jimeng Sun. PILOT: Legal case outcome prediction with case law. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 609–621, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.
- [3] Dinh-Truong Do, Son T. Luu, Trang Pham, Trung Vo, Nguyen-Hoang Chu, Quang-Huy Chu, Cuong Nguyen, Minh Nguyen, An Trieu, Dat Nguyen, et al. A summary of the ALQAC 2024 competition. In *Proceedings of the 2024 16th International Conference on Knowledge and Systems Engineering*, pages 422–427. IEEE, 2024. doi: 10.1109/KSE63888.2024.11063484.
- [4] Dinh-Truong Do, Son T. Luu, Trang Pham, Trung Vo, Nguyen-Hoang Chu, Quang-Huy Chu, Cuong Hoang, Tan-Minh Nguyen, An Trieu, Dat Nguyen, et al. A summary of the ALQAC 2025 competition. In *Proceedings of the 2025 17th International Conference on Knowledge and Systems Engineering*, pages 1–6. IEEE, 2025. doi: 10.1109/KSE68178.2025.11309650.
- [5] Huy D. T. Do, Hien D. Nguyen, Vuong T. Pham, and Tri-Hai Nguyen. Legal-onto model for efficient land law updates in vietnam. In *Future Data and Security Engineering. Big Data, Security and Privacy, Smart City and Industry 4.0 Applications*, pages 241–253. Springer, 2024. doi: 10.1007/978-981-96-0434-0_17.
- [6] Tinh Pham Phuc Do, Ngoc Dinh Duy Cao, Khanh Quoc Tran, and Kiet Van Nguyen. R2GQA: Retriever-reader-generator question answering system to support students understanding legal regulations in higher education. *Artificial Intelligence and Law*, 2025. doi: 10.1007/s10506-025-09457-7.
- [7] Nhung Thi-Hong Duong, Mai Ngoc Ho, Tin Van Huynh, and Kiet Van Nguyen. ViLegalNLI: Natural language inference for vietnamese legal texts. *arXiv preprint arXiv:2605.00116*, 2026. doi: 10.48550/arXiv.2605.00116.
- [8] Kazuma Kadowaki and Yoshinobu Kano. Japanese legal judgment prediction using ModernBERT and generative AI-based information extraction. *The Review of Socionetwork Strategies*, 20:313–339, 2026. doi: 10.1007/s12626-026-00201-4.
- [9] Phi Manh Kien, Ha-Thanh Nguyen, Ngo Xuan Bach, Vu Tran, Minh Le Nguyen, and Tu Minh Phuong. Answering legal questions by learning neural attentive text representation. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 988–998, Barcelona, Spain (Online), December 2020. International Committee on Computational Linguistics. doi: 10.18653/v1/2020.coling-main.86. URL <https://aclanthology.org/2020.coling-main.86/>.
- [10] Arooba Maqsood, Adnan Ul-Hasan, and Faisal Shafait. Transformer-based architecture for judgment prediction and explanation in legal proceedings. In *Document*

Analysis Systems, volume 14994 of *Lecture Notes in Computer Science*, pages 20–36. Springer, 2024. doi: 10.1007/978-3-031-70442-0_2.

- [11] National Assembly of Vietnam. Quoc hoi nghe bao cao ve ket qua cong tac nam 2025 cua nganh toa an nhan dan. <https://quochoi.vn/tintuc/Pages/tin-hoat-dong-cua-quoc-hoi.aspx?ItemID=97445>, 2025. Accessed: 2026-09-12.
- [12] Chau Nguyen, Minh-Quan Bui, Dinh-Truong Do, Nguyen-Khang Le, Dieu-Hien Nguyen, Thu-Trang Nguyen, Ha-Thanh Nguyen, Vu Tran, Le-Minh Nguyen, et al. ALQAC 2022: A summary of the competition. In *Proceedings of the 2022 14th International Conference on Knowledge and Systems Engineering*, pages 1–5. IEEE, 2022. doi: 10.1109/KSE56063.2022.9953764.
- [13] Chau Nguyen, Son T. Luu, Thanh Tran, An Trieu, Anh Dang, Dat Nguyen, Hiep Nguyen, Tin Pham, Trang Pham, Thien-Trung Vo, et al. A summary of the ALQAC 2023 competition. In *Proceedings of the 2023 15th International Conference on Knowledge and Systems Engineering*, pages 1–6. IEEE, 2023. doi: 10.1109/KSE59128.2023.10299527.
- [14] Chau Nguyen, Thanh Tran, Khang Le, Hien Nguyen, Truong Do, Trang Pham, Son T. Luu, Trung Vo, and Le-Minh Nguyen. Pushing the boundaries of legal information processing with integration of large language models. In *New Frontiers in Artificial Intelligence*, pages 167–182. Springer, 2024. doi: 10.1007/978-981-97-3076-6_12.
- [15] Dat Nguyen, Minh-Phuong Nguyen, Quang-Huy Chu, Son T. Luu, Nguyen-Hoang Chu, Trung Vo, and Le-Minh Nguyen. Enhancing legal text processing and structural analysis with large language models at COLIEE 2025. *The Review of Socionetwork Strategies*, 20(1):361–383, 2026. doi: 10.1007/s12626-026-00211-2.
- [16] Ha Thanh Nguyen, Minh Quan Bui, Chau Nguyen, Tung Le, Minh Phuong Nguyen, Tran Binh Dang, Thi Hai Yen Vuong, Teerada Rajacharak, Le Minh Nguyen, et al. A summary of the ALQAC 2021 competition. In *Proceedings of the 2021 13th International Conference on Knowledge and Systems Engineering*, pages 1–6. IEEE, 2021. doi: 10.1109/KSE53942.2021.9648724.
- [17] Hien D. Nguyen, Vuong T. Pham, Quoc Hung Ngo, Anh T. Huynh, Binh T. Nguyen, and Jose Machado. Intelligent search system for resume and labor law. *PeerJ Computer Science*, 10:e1786, 2024. doi: 10.7717/peerj-cs.1786.
- [18] Quan Van Nguyen, Anh Trong Nguyen, Huy Quang Pham, and Kiet Van Nguyen. Pre-trained language models fine-tuned with SVM for legal textual entailment recognition. *Journal of Computer Science and Cybernetics*, 41(3):305–321, 2025. doi: 10.15625/1813-9663/20618.
- [19] Tan-Minh Nguyen, Hoang-Trung Nguyen, Trong-Khoi Dao, Xuan-Hieu Phan, Ha-Thanh Nguyen, and Thi-Hai-Yen Vuong. VLQA: The first comprehensive, large, and high-quality vietnamese dataset for legal question answering. *arXiv preprint arXiv:2507.19995*, 2025. doi: 10.48550/arXiv.2507.19995.
- [20] Shubham Kumar Nigam, Aniket Deroy, Subhankar Maity, and Arnab Bhattacharya. Rethinking legal judgement prediction in a realistic scenario in the era of large language models. In *Proceedings of the Natural Legal Language Processing Workshop 2024*, pages 61–80, Miami, FL, USA, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.nllp-1.6. URL <https://aclanthology.org/2024.nllp-1.6/>.
- [21] Shubham Kumar Nigam, Deepak Patnaik Balaramamahanthi, Shivam Mishra, Noel Shallum, Kripabandhu Ghosh, and Arnab Bhattacharya. NyayaAnumana and INLegalLlama: The largest Indian legal judgment prediction dataset and specialized language model for enhanced decision analysis. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 11135–11160, Abu Dhabi, UAE, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.coling-main.738/>.
- [22] Joel Niklaus, Ilias Chalkidis, and Matthias Stürmer. Swiss-judgment-prediction: A multilingual legal judgment prediction benchmark. In *Proceedings of the Natural Legal Language Processing Workshop 2021*, pages 19–35, Punta Cana, Dominican Republic, 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.nllp-1.3. URL <https://aclanthology.org/2021.nllp-1.3/>.
- [23] Vuong Pham, Hoang Huy Le, Thinh Phu Ngo, Binh Nguyen, Diem Nguyen, and Hien D. Nguyen. Enhancing legal research through knowledge-infused information retrieval for vietnamese labor law. *IAES International Journal of Artificial Intelligence*, 13(4):3962–3973, 2024. doi: 10.11591/ijai.v13.i4.pp3962-3973.
- [24] Ruihao Shui, Yixin Cao, Xiang Wang, and Tat-Seng Chua. A comprehensive evaluation of large language models on legal judgment prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 7337–7348, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.

490. URL <https://aclanthology.org/2023.findings-emnlp.490/>.

- [25] Jingyun Sun, Shaobin Huang, and Chi Wei. Chinese legal judgment prediction via knowledgeable prompt learning. *Expert Systems with Applications*, 238: 122177, 2024. doi: 10.1016/j.eswa.2023.122177.
- [26] Dietrich Trautmann, Alina Petrova, and Frank Schilder. Legal prompt engineering for multilingual legal judgment prediction. *arXiv preprint arXiv:2212.02199*, 2022. doi: 10.48550/arXiv.2212.02199.
- [27] Chaojun Xiao, Haoxi Zhong, Zhipeng Guo, Cunchao Tu, Zhiyuan Liu, Maosong Sun, Yansong Feng, Xi-anpei Han, Zhen Hu, Heng Wang, and Jianfeng Xu. CAIL2018: A large-scale legal dataset for judgment prediction. *arXiv preprint arXiv:1807.02478*, 2018. doi: 10.48550/arXiv.1807.02478.
- [28] Yue Zhang, Zhiliang Tian, Shicheng Zhou, Haiyang Wang, Wenqing Hou, Yuying Liu, Xuechen Zhao, Minlie Huang, Ye Wang, and Bin Zhou. RLJP: Legal judgment prediction via first-order logic rule-enhanced with large language models. *arXiv preprint arXiv:2505.21281*, 2025. doi: 10.48550/arXiv.2505.21281.